

Relationship Types & Relationship Sets

Q. Explain the concepts of Relationship Types and Relationship Sets. Discuss the Degree of Relationship and Mapping Cardinalities (1:1, 1:N, M:N) with examples.

> **Definition to Remember**

> A **diagram**
diagram
moment

1. Relationship Type and Relationship Set

- **Relationship Type:** Association between entity types. E.g., `WORKS_FOR` links `EMPLOYEE` and `DEPARTMENT`.

- **R**

2. Degree of Relationship

The **Degree** = number of entity types participating in the relationship.

| Degree | Name | Example |

| :--- | :

3. Mapping Cardinalities

Mapping Cardinality defines the maximum number of relationship instances an entity can participate in.

1. One-to-One (1:1)

- On

2. One-to-Many (1:N)

- On

3. Many-to-One (N:1)

- Mu

4. Many-to-Many (M:N)

- Ent

1:1 A

4. Participation Constraints

| Type | Description | ER Symbol |

- **Total Example:** Every `EMPLOYEE` must belong to a `DEPARTMENT`.

> **Must-Write Points (for 10 marks)**

> **Quick Recall**